

Introductory
Night Sky
photography

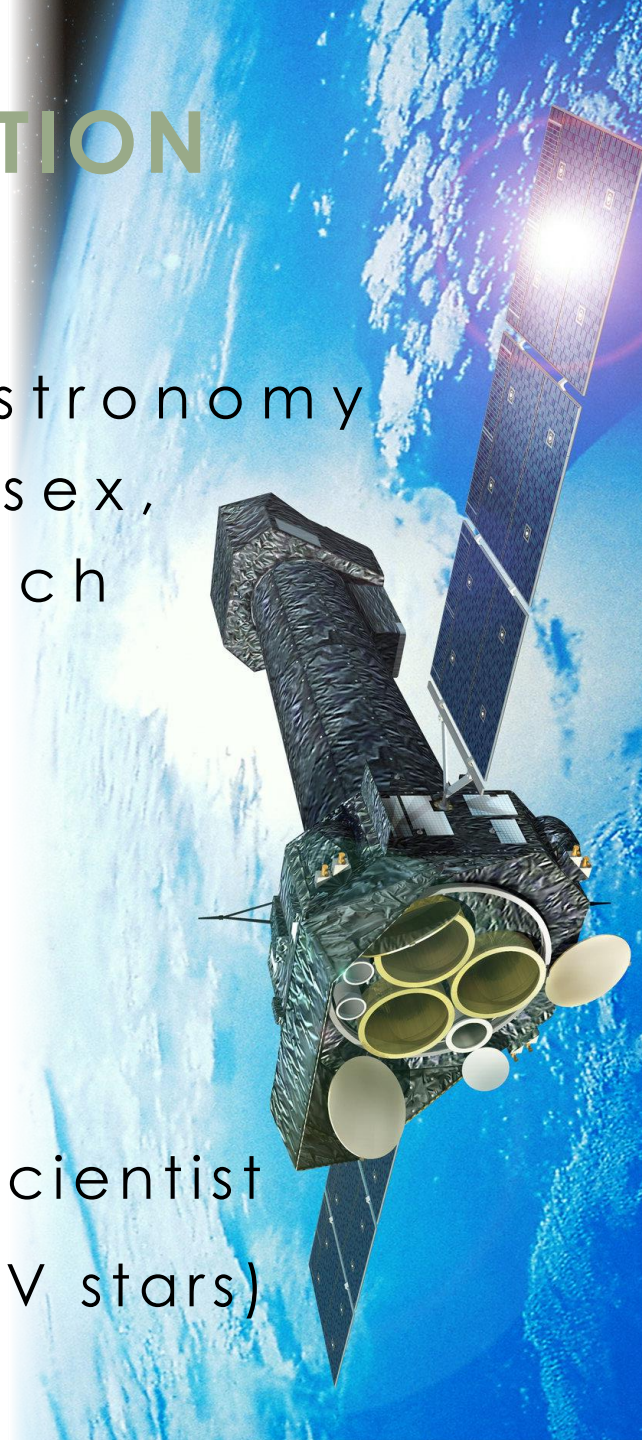
Dr Darren (Das) Baskill
University of Sussex

WELCOME & INTRODUCTION

Lecturer in Physics & Astronomy
at the University of Sussex,
also running our outreach
programme.

Previous jobs include:

- Educator at the Royal
Observatory Greenwich
- XMM-Newton calibration scientist
- PhD in X-ray astronomy (CV stars)





Observing in Hawai'i...



Doing outreach in schools & elsewhere

MEDIA WORK



THE CONVERSATION

Academic rigour, journalistic flair

Search analysis, research, academics...

Arts • Culture • Business • Economy • Education • Environment • Health • Politics • Society • Science • Tech • World • Podcasts • Insights

Why do the aurora look better through a camera? And how do you spot a fake image? Q&A with an astrophotographer

Published: May 14, 2024 2:37pm BST

Aurora spotting in Kendal, northern England, May 10, 2024. Stop Press Media/Alamy

On 10 and 11 May 2024, large parts of the world were treated to their most spectacular display of the aurora – the northern and southern lights – in a generation. Thanks to modern cameras, the phenomenon was all over social media. It seems almost everyone knew someone who

Author

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Astrophotography

1. The Sun & Moon (& settings)

2. Wide angle night sky

3. Processing &
Deep sky targets
with a telephoto lens

4. Time-lapse photography

Timings

10:30-11:15 The Sun & Moon

11:30-12:45 Wide angle night sky

13:45-14:30 Processing &

Deep sky targets

14:45-15:45 Time-lapse photography



Imaging the Sun & Moon



The Sun



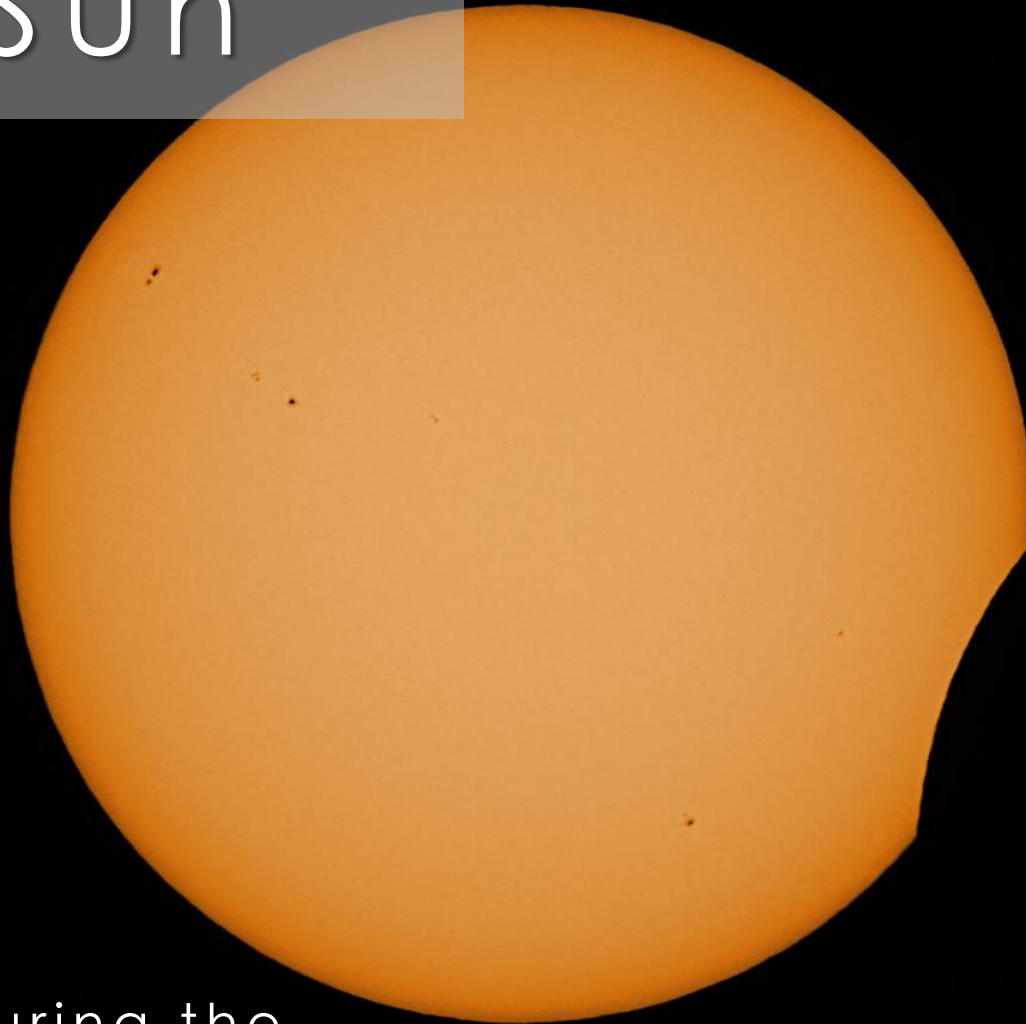
While the setting Sun is faint enough for a camera, due to atmospheric absorption...

The Sun



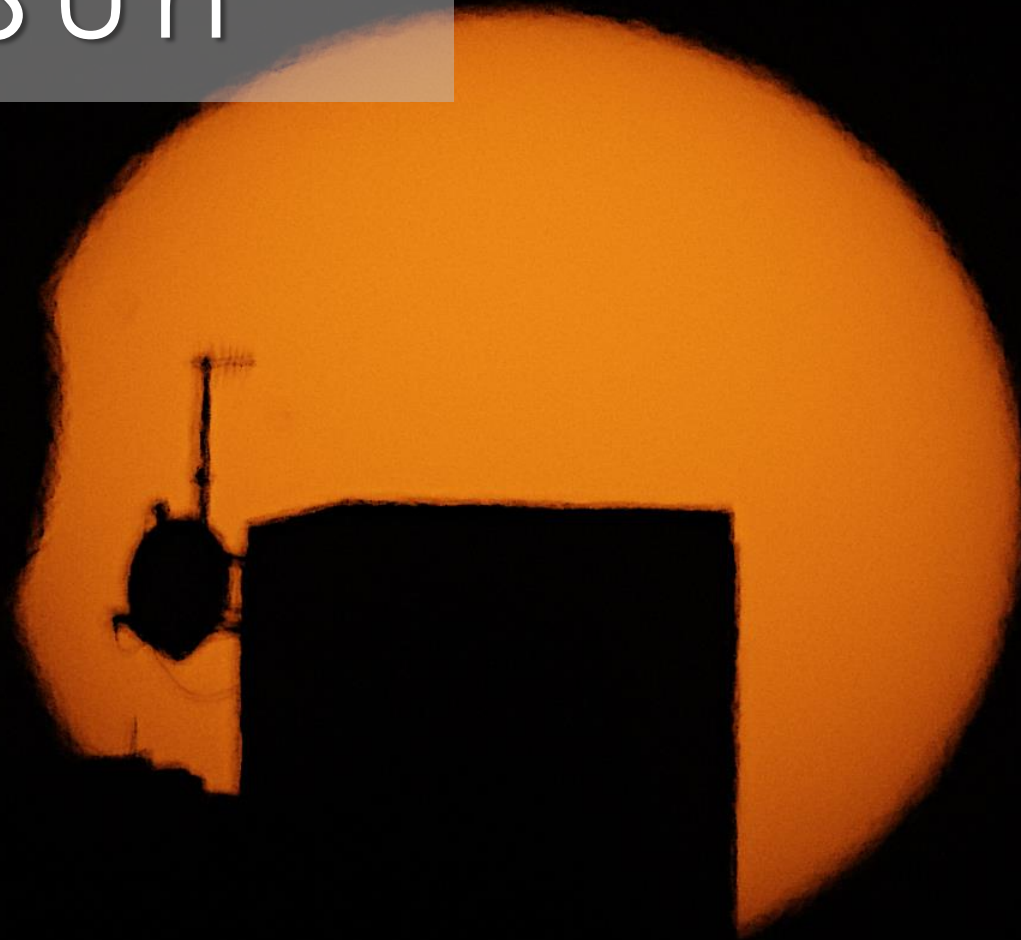
A solar filter is usually required for solar photography to prevent damaging your camera

The Sun



As I used during the
solar eclipse of the 12th August, 2026.

The Sun



As I used during the
solar eclipse of the 12th August, 2026.

The Sun

Next partial
solar eclipse
50% from the UK

~9-11am
2nd August 2027



The Sun



Pin hole
projection
with...

Coffee cups

Scarfs

Colander

Card + holes

Hands

Anything!

The Sun

Specialised solar
telescopes are available



The Sun



While the Sun
looks a strange colour
through their filters...

The Sun



This colour can be
changed with editing software.

The Sun & Moon

Method

- Autofocus
- Autoexposure
- Auto f/number



The Moon



The Moon



The Moon



The Moon

A night photograph of the Tower Bridge in London. The bridge's two towers are illuminated with warm yellow lights, and the suspension cables are lit with blue. A large, bright, orange-hued full moon is visible in the dark blue sky to the right of the bridge. In the background, other city buildings are visible, some with lights on. The overall scene is a composite image where the moon's position is planned using software.

Planned using Stellarium
and an OS map

Method: Camera settings

- f/number
- Exposure time
- ISO setting

Method: Camera settings

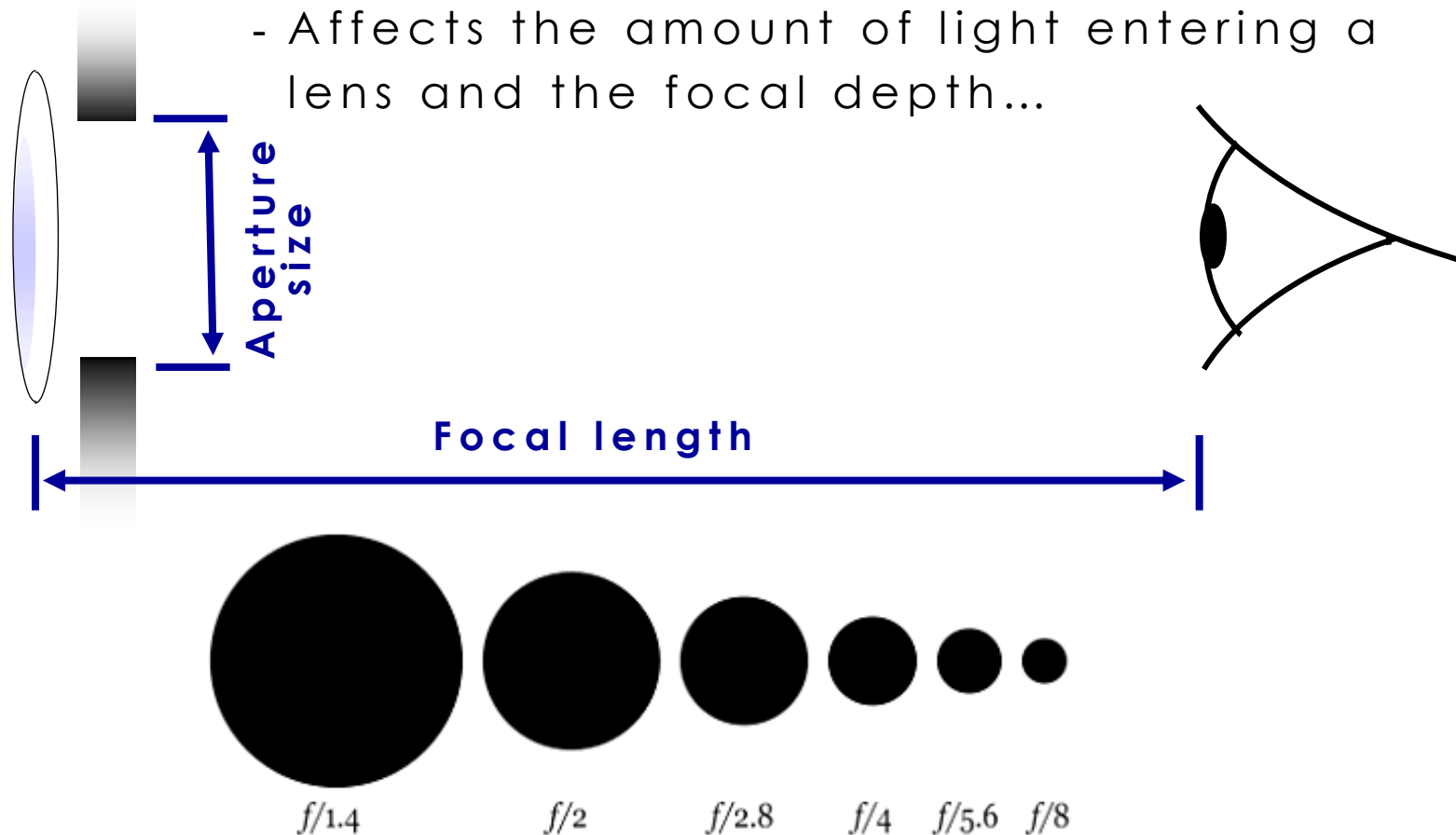
- f/number
 - Exposure time
 - ISO setting
- 
- Decrease to collect more light...
- ... Or increase to get a greater depth of field.

f-number

f/number is the aperture size
(opening) of a lens,
in terms of focal length

- f/2 means that the aperture (opening) is one half-the focal length of the lens

- Affects the amount of light entering a lens and the focal depth...



$f/4.5$



$f/6.7$



$f/9.5$



f/13



f/32





f/5



f/32



The Moon

By Laurent Laveder

<http://www.pixheaven.net>



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<http://www.pixheaven.net>

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<http://www.pixheaven.net>



f/5.6



The Moon

f/11



Diffraction
spikes

The Moon





The Moon

f/20



The Moon

f/20

Consider foreground
objects for an interesting
composition



The Moon



The Moon



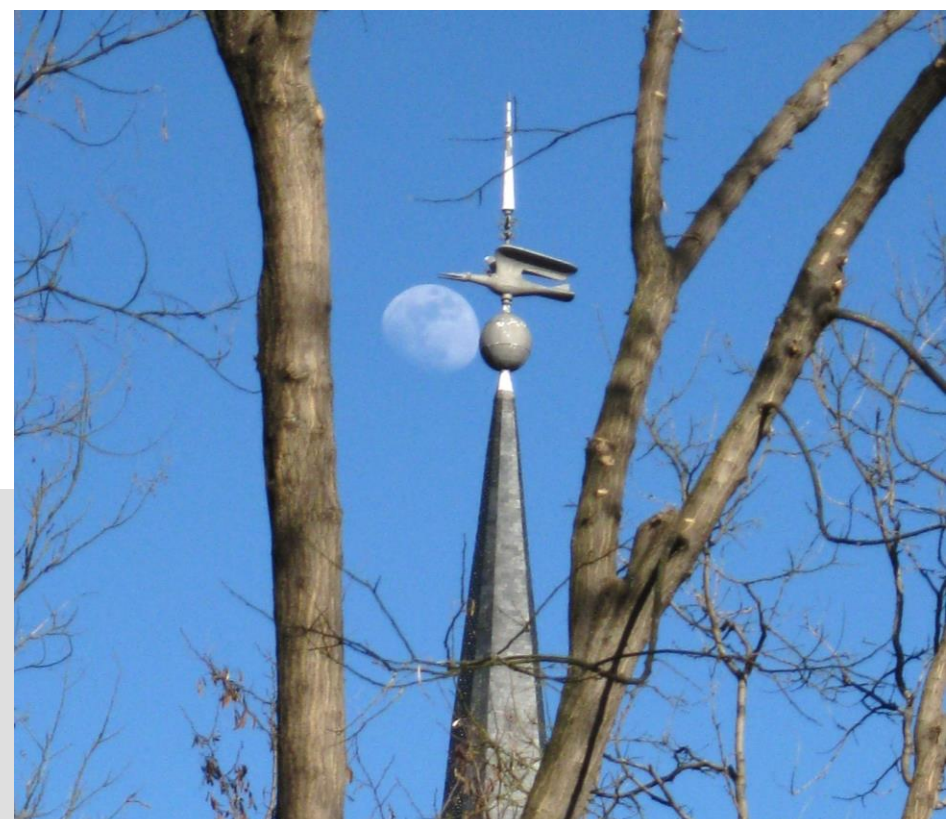
The Moon



The Moon



The Moon



The Moon
over London,
Vienna & Madrid

Camera adapters





The Moon

The Sun & Moon Summary

- Always search
for interesting
foreground objects
- Focal depth?
Diffraction spikes?



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